

Aggregate Expenditure and Output in the Short Run

ECON 101 – Principles of Macroeconomics

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- 8.1 The Aggregate Expenditure Model
- 8.2 Determining Aggregate Expenditure in the Economy
- 8.3 Graphing Macroeconomic Equilibrium
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8.1 Learning Objective

Goal

Understand how macroeconomic equilibrium is determined in the aggregate expenditure model.

- We build the **aggregate expenditure (AE) model**: a short-run model.
- It studies the relationship between total spending and output,
- **assuming the price level is constant** in the short run.
- Why study this? It helps us understand:
 - why recessions occur,
 - why government stimulus can boost the economy,
 - and how the **multiplier effect** works.

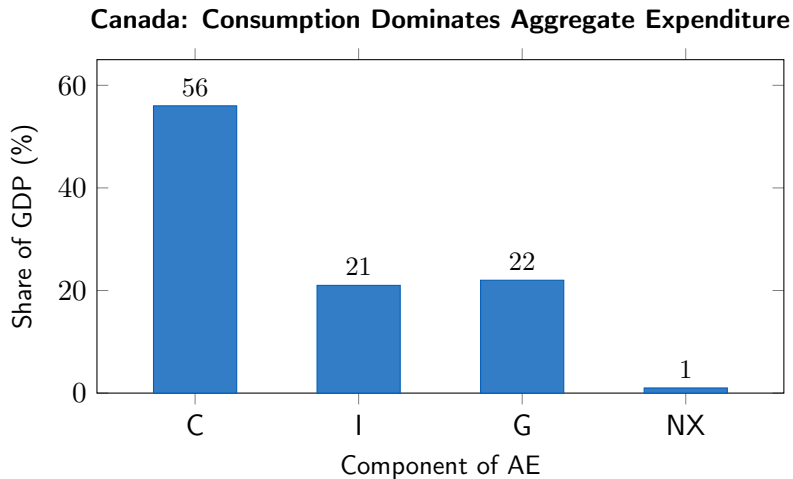
Aggregate Expenditure: Definition

- **Aggregate expenditure (AE):** total planned spending in the economy.

$$AE = C + I + G + NX$$

- C : consumption (households buying groceries, cars, Netflix subscriptions)
- I : planned investment (firms buying machines, households buying new homes)
- G : government purchases (federal, provincial, municipal)
- NX : net exports (exports minus imports)
- **Canadian AE composition (2023, Statistics Canada):**
 - $C \approx 56\%$, $I \approx 21\%$, $G \approx 22\%$, $NX \approx 1\%$ of GDP.

Table 8.1 – Canadian Components of AE (2023)



Source: Statistics Canada, National Accounts (Table 36-10-0104-01), 2023 estimates.

Planned vs. Actual Investment

- National accounts measure **actual investment**.
- AE model uses **planned investment**: what firms intend to invest.
- **Actual investment** = planned investment + **unplanned inventory changes**.
- **Why inventories matter**:
 - If sales are **lower than expected**: inventories build up unintentionally (firms reduce output).
 - If sales are **higher than expected**: inventories fall unintentionally (firms increase output).
- **Canadian example (2022)**: supply chain disruptions left Canadian auto dealers with record-low inventories — firms scrambled to increase output.

Macroeconomic Equilibrium in the AE Model

- **Macroeconomic equilibrium:**

$$\text{Planned AE} = \text{Real GDP}$$

- If planned AE > output:
 - firms sell more than expected, inventories fall,
 - firms **increase** production and employment.
- If planned AE < output:
 - firms sell less than expected, inventories rise,
 - firms **reduce** production and employment.
- **Inventories are the adjustment mechanism** that pushes the economy toward equilibrium.

Determinants of Consumption

Increase Consumption

- Higher disposable income
- Higher household wealth (house prices, stock market)
- Higher expected future income
- Lower interest rates
- Higher consumer confidence

Decrease Consumption

- Lower disposable income
 - Wealth destruction (market crash)
 - Pessimistic income expectations
 - Higher interest rates
 - Low consumer confidence
-
- **Canadian example (2022–2024):** Bank of Canada raised overnight rate from 0.25% to 5.0%. This reduced consumer spending, especially on mortgages, autos, and durables — Canadian consumer confidence fell sharply.

The Consumption Function and MPC

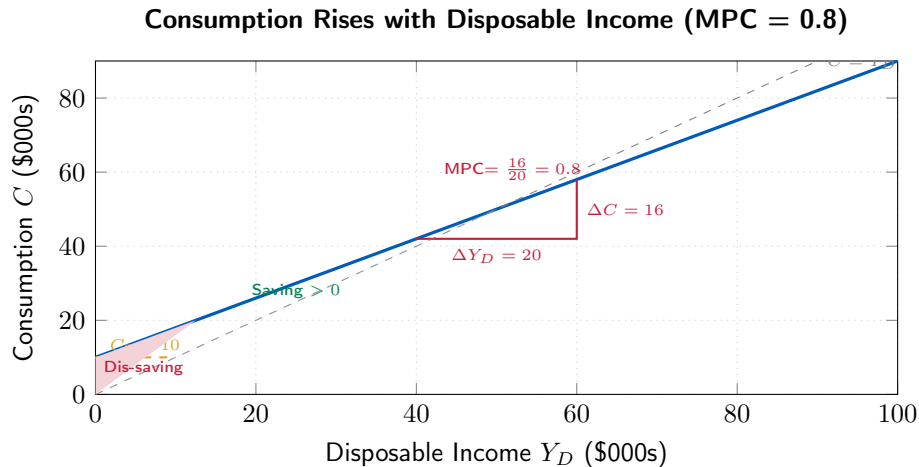
- The **consumption function**:

$$C = C_0 + \text{MPC} \times Y_D,$$

where:

- C_0 = autonomous consumption (consumption when $Y_D = 0$),
 - Y_D = disposable income = $Y - T_{\text{net}}$,
 - MPC = marginal propensity to consume.
- **MPC:**
$$\text{MPC} = \frac{\Delta C}{\Delta Y_D}.$$
 - **Canadian estimate (Statistics Canada):** $\text{MPC} \approx 0.75\text{--}0.80$.
 - A \$1,000 tax rebate $\Rightarrow$ roughly **\$750–\$800** in new spending.

Figure 8.1 – The Consumption Function



- Marginal propensity to consume (MPC):

$$MPC = \frac{\Delta C}{\Delta Y_D}$$

- Marginal propensity to save (MPS):

$$MPS = \frac{\Delta S}{\Delta Y_D} = 1 - MPC$$

- Every extra dollar of income is either spent or saved:

$$MPC + MPS = 1.$$

- **Example:** $MPC = 0.8$, so $MPS = 0.2$.
 - Of an extra \$1,000 income: \$800 spent, \$200 saved.

Determinants of Planned Investment

- **Expectations of future profits:**
 - Firms invest when optimistic. Recessions reduce planned investment.
 - **Canada 2020:** business investment fell $\approx 15\%$ due to pandemic uncertainty.
- **Real interest rate:**
 - Higher rates $\Rightarrow$ higher cost of borrowing $\Rightarrow$ less investment.
 - **Canada 2022–23:** BoC hikes pushed mortgage rates to 6–7%, residential construction fell sharply.
- **Taxes:**
 - Higher corporate tax rates reduce after-tax returns on investment.
 - Canada's federal corporate rate: 15% (base) + provincial rate.
- **Cash flow:**
 - Many firms finance investment internally. Lower profits $\Rightarrow$ less cash $\Rightarrow$ less investment.

Determinants of Net Exports (NX)

- **Net exports** = exports – imports.
- **Determinants:**
 - **Relative price levels:** if Canadian prices rise faster than foreign prices, exports fall and imports rise $\Rightarrow$ NX falls.
 - **Relative growth rates:** if Canada grows faster than trading partners, imports rise $\Rightarrow$ NX falls.
 - **Exchange rate (CAD/USD):**
 - **CAD appreciates** (CAD strengthens): Canadian exports more expensive for foreigners; imports cheaper $\Rightarrow$ NX falls.
 - **CAD depreciates** (CAD weakens): Canadian exports cheaper; imports more expensive $\Rightarrow$ NX rises.
- **Canada's key trading partner:** the U.S. takes $\approx 75\%$ of Canadian exports (Statistics Canada, 2024).

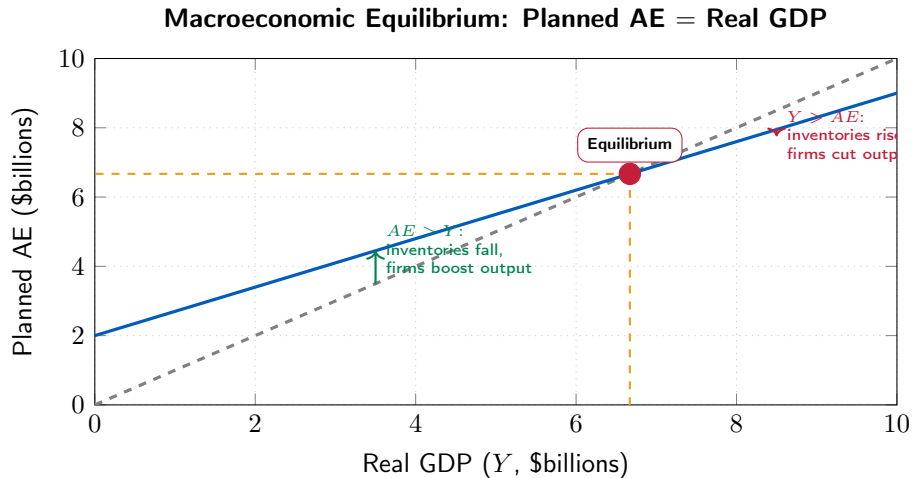
8.3 Learning Objective

Goal

Use a 45°-line diagram to illustrate macroeconomic equilibrium.

- Graph **real GDP** on the horizontal axis.
- Graph **planned aggregate expenditure** on the vertical axis.
- The **45° line** shows all points where $AE = Y$ (equilibrium).
- The AE line is **upward sloping** with slope = MPC (since higher $Y \Rightarrow$ higher $C \Rightarrow$ higher AE).

Figure 8.2 – The 45°-Line Diagram



Understanding the 45°Line

- **On the 45°line:** Real GDP = Planned AE — equilibrium.
- **Above 45°line** ($AE > Y$):
 - Planned spending exceeds output.
 - Inventories are depleted.
 - Firms **increase** production $\Rightarrow$ GDP rises toward equilibrium.
 - **Example:** post-COVID reopening 2021 — consumers spent more than firms produced, pushing GDP higher.
- **Below 45°line** ($AE < Y$):
 - Planned spending is less than output.
 - Inventories accumulate unintentionally.
 - Firms **reduce** production $\Rightarrow$ GDP falls toward equilibrium.
 - **Example:** 2020 COVID lockdowns — businesses produced but few consumers could buy.

8.4 Learning Objective

Goal

Describe the multiplier effect and use the multiplier formula to calculate changes in equilibrium GDP.

- A small change in **autonomous expenditure** causes a **larger** change in equilibrium real GDP.
- This amplification is called the **multiplier effect**.
- Why? Because an initial increase in spending raises incomes, which raises consumption, which raises incomes further, and so on.

The Multiplier: Step-by-Step

- Suppose the federal government spends **\$10 billion** on new broadband infrastructure.
- **Round 1:** Telecom workers and contractors receive \$10B income.
 - They spend $MPC \times 10 = 0.8 \times 10 = \$8B$ on goods and services.
- **Round 2:** Retailers and service providers receive \$8B.
 - They spend $0.8 \times 8 = \$6.4B$.
- **Round 3:** $0.8 \times 6.4 = \$5.12B$, and so on...
- **Total effect:** $\Delta Y = 10 + 8 + 6.4 + 5.12 + \dots = \frac{1}{1-0.8} \times 10 = \mathbf{\$50B}$.

Figure 8.3 – The Multiplier Effect (MPC = 0.8)

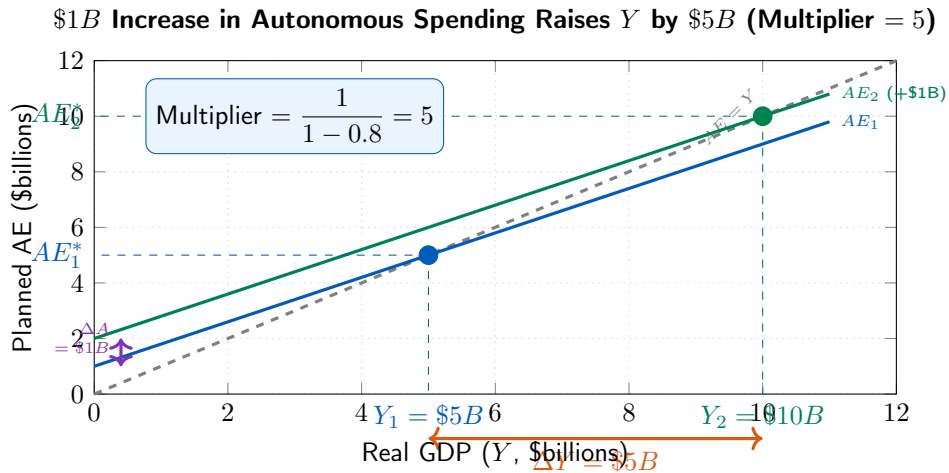
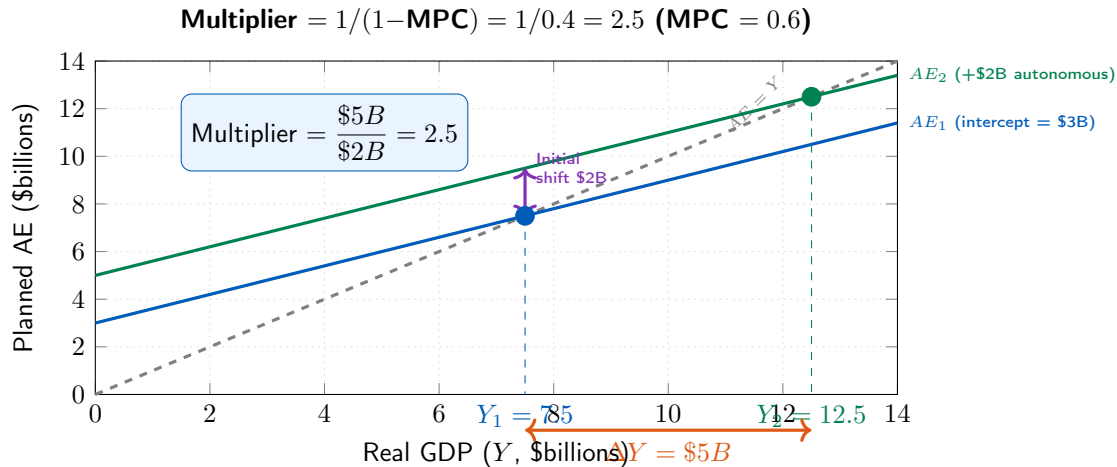


Figure 8.3b – Multiplier Effect (Cleaner Illustration)



Formula for the Multiplier

- The total change in equilibrium GDP from an initial change ΔA in autonomous expenditure:

$$\Delta Y = \Delta A [1 + \text{MPC} + \text{MPC}^2 + \text{MPC}^3 + \dots] = \frac{1}{1 - \text{MPC}} \times \Delta A.$$

- Multiplier:**

$$\text{Multiplier} = \frac{1}{1 - \text{MPC}} = \frac{1}{\text{MPS}}$$

- Examples:**

- MPC = 0.8: Multiplier = $\frac{1}{0.2} = 5$.
 - MPC = 0.75: Multiplier = $\frac{1}{0.25} = 4$.
 - MPC = 0.5: Multiplier = $\frac{1}{0.5} = 2$.
- Real-world multiplier:** much smaller (1.0–1.5) due to taxes, imports, and price changes.

The Paradox of Thrift

- **Long run:** higher saving $\Rightarrow$ more investment $\Rightarrow$ higher growth.
- **Short run (AE model):** if all households simultaneously save more (consume less):
 - Consumption falls, AE falls, equilibrium GDP falls.
 - Could trigger a recession — even though each individual was being prudent.
- This is the **paradox of thrift**: what is good for individuals can be bad for the economy in the short run.
- **Canadian example (2020):** household saving rate surged to **28%** during lockdowns.
 - GDP fell sharply (demand shock).
 - The “forced saving” was then partially spent during the 2021–22 reopening boom.

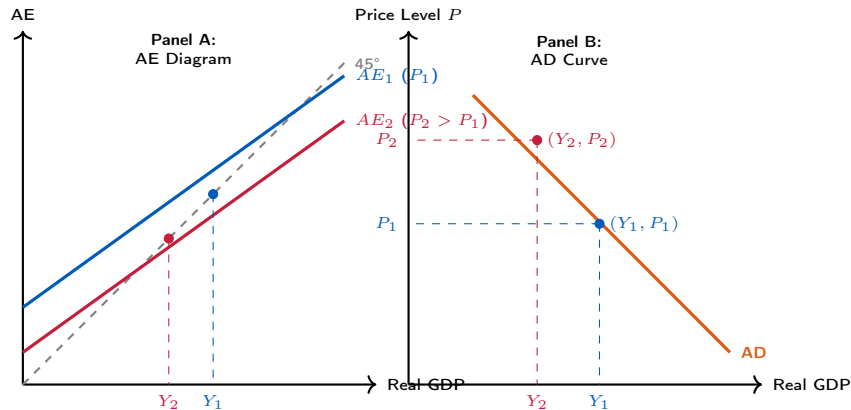
8.5 Learning Objective

Goal

Understand the relationship between the aggregate demand curve and aggregate expenditure.

- So far: price level fixed.
- Now: what happens to AE when the **price level changes**?
- Three channels link a higher price level to lower AE:
 1. **Wealth effect** $\Rightarrow$ lower C .
 2. **Interest-rate effect** $\Rightarrow$ lower I .
 3. **International-trade effect** $\Rightarrow$ lower NX .
- A higher price level $\Rightarrow$ lower equilibrium GDP $\Rightarrow$ downward-sloping **aggregate demand curve**.

Figure 8.4 – From AE to the Aggregate Demand Curve



Higher price level $\Rightarrow$ AE shifts down $\Rightarrow$ lower equilibrium Y $\Rightarrow$ downward-sloping AD.

Appendix – Solving the AE Model Algebraically

- Suppose the Canadian economy is described by:

$$\begin{aligned}C &= 400 + 0.8(Y - T), \\I &= 300, \quad G = 500, \quad T = 250, \quad NX = -50.\end{aligned}$$

- Equilibrium condition: $Y = C + I + G + NX$.
- Substituting:

$$Y = 400 + 0.8(Y - 250) + 300 + 500 - 50.$$

$$Y = 400 + 0.8Y - 200 + 750 \quad \Rightarrow \quad 0.2Y = 950 \quad \Rightarrow \quad \boxed{Y = \$4,750 \text{ billion}}.$$

- General multiplier: $\frac{1}{1-\text{MPC}} = \frac{1}{1-0.8} = 5$.

- The **AE model** determines short-run equilibrium when the price level is fixed.
- Equilibrium: $\text{Planned AE} = Y$; inventories adjust to reach this point.
- **Consumption** driven by disposable income ($\text{MPC} \approx 0.75\text{--}0.80$ in Canada), wealth, expectations, and interest rates.
- **Planned investment** driven by profit expectations, interest rates, taxes, and cash flow.
- **Net exports** driven by relative prices, relative growth rates, and the exchange rate.
- The **multiplier** $= 1/(1 - \text{MPC})$ amplifies changes in autonomous spending:
 - Works in both directions — stimulus boosts GDP, contractionary shock reduces it.
 - Real-world multipliers are smaller due to taxes, imports, and price changes.
- The **paradox of thrift**: individually rational saving can reduce GDP in the short run.
- As the price level changes, AE shifts, tracing out the downward-sloping **aggregate demand curve**.

Thank you!

Questions?